Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$M_{0c}^{a}M_{0b}^{c} = \left(M_{0}^{2}\right)_{b}^{a} = \begin{pmatrix} -\frac{\alpha'}{2}\left[A_{\mu},A_{\nu}\right]\psi^{\mu}\psi^{\nu} + \overline{T}T & -i\sqrt{\alpha'}\left(\overline{T}A'_{\mu} - A_{\mu}\overline{T}\right)\psi^{\mu} \\ -i\sqrt{\alpha'}\left(TA_{\mu} - A'_{\mu}T\right)\psi^{\mu} & -\frac{\alpha'}{2}\left[A_{\mu'},A'_{\nu}\right]\psi^{\mu}\psi^{\nu} + T\overline{T} \end{pmatrix}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \|r_h^+u - r_hu\|_{0,2}^2 &= a_h^+(r_h^+u - r_hu, w) \\ &= a_h^+(r_h^+u - r_hu, w - w_h^+) + a_h^+(r_h^+u - r_hu, w_h^+) \\ &= a_h^+(r_h^+u - r_hu, w - w_h^+) + a_h^+(u - r_hu, w_h^+) \\ &= a_h^+(r_h^+u - r_hu, w - w_h^+) + a_h^+(u - r_hu, w_h^+ - w_h) \\ &\quad + a_h^+(u - r_hu, w_h) - a_{h(u-r_hu,w_h)} \\ &=: T_1 + T_2 + T_3, \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sum_{n=1}^{\infty} c_{1,n}T_1^n\left(\left\{x_\tau^0\right\}_{\tau=1}^{r}\right) &\equiv \sum_{n=1}^{\infty} c_{1,n}T_1^n\left(\left\{x_\tau^0\tilde{f}_0^n\right\}_{\tau=1}^{r}\right), \\ \sum_{n=1}^{\infty} c_{2,n}T_2^n\left(\left\{x_\theta^0\right\}_{\theta=r+1}^{r+s}\right) &\equiv \sum_{n=1}^{\infty} c_{2,n}T_2^n\left(\left\{x_\theta^0\tilde{f}_0^n\right\}_{\theta=r+1}^{r+s}\right). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\|\psi_{n,t}(b)^*\psi_{n,t}(b) + \left(\psi_{n,e}(c) + \alpha 1_{k_n}\right)^*\left(\psi_{n,e}(c) + \alpha 1_{k_n}\right) - \|b\|^2\ 1_{k_n}\| \\ &\qquad \le \|\psi_{n,t}(b)^*\psi_{n,t}(b) - \psi_{n,e}(b^*b)\| + \|\psi_{n,e}(c)^*\psi_{n,e}(c) - \psi_{n,e}(c^*c)\| \\ &\qquad = \|\psi_{n,t^{-1}}(b^*)\psi_{n,t}(b) - \psi_{n,e}(b^*b)\| + \|\psi_{n,e}(c^*)\psi_{n,e}(c) - \psi_{n,e}(c^*c)\| < 2\varepsilon_n \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\text{(i)}\quad B = B_1 \cup B_2, \tilde{B} = E_1 \cup E_2 \ \text{ and } \ \omega(E_i) \ge \frac{1}{2}\omega\big(\tilde{B}\big), i = 1, 2; \\ &\text{(ii)}\quad b(x) - b(y) \ \text{ does not change sign for all } \ (x, y) \in B_i \times E_i, i = 1, 2; \\ &\text{(iii)}\quad |b(x) - m_{b(\tilde{B})}| \le |b(x) - b(y)| \ \text{ for all } \ (x, y) \in B_i \times E_i, i = 1, 2. \end{aligned}$$